Results in Applied Mathematics 21 (2024) 100418

Contents lists available at ScienceDirect

Results in Applied Mathematics

journal homepage: www.elsevier.com/locate/results-in-applied-mathematics

A generalized finite difference method for 2D dynamic crack analysis

Bingrui Ju [a,*], Boyang Yu [b], Zhiyuan Zhou [c]

[a] School of Mathematics and Statistics, Qingdao University, Qingdao, 266071, PR China
[b] Qingdao, 266071, PR China
[c] Viterbi School of Engineering, University of Southern California, Los Angeles, CA, 90015, USA

ARTICLE INFO

Keywords:
Dynamic crack analysis
Meshless method
Generalized finite difference method
Dynamic stress intensity factor
Fracture mechanics

ABSTRACT

This paper presents a new framework for efficient and accurate analysis of transient elastodynamic cracks by using the generalized finite difference method (GFDM). The method first discretizes the solution domain into a set of overlapping small subdomains, and then in each of the subdomains, the unknown functions and their derivatives are approximated by using the local Taylor series expansions and moving-least square approximation. The degree of the Taylor series used in the local subdomain is increased automatically in the regions near the crack-tips, in order to appropriately describe the local asymptotic behavior of near-tip displacement and stress fields. The path-independent J-integral and sub-domain technique are adopted to compute the dynamic stress intensity factors (SIFs) of the cracked bodies. Preliminary numerical experiments for dynamic SIFs with both uniform and variable loading conditions are given to show the efficient and accuracy of the present method for transient elastodynamic crack analysis.

1. Introduction

An accurate and efficient analysis of the dynamic fracture behavior of a cracked body is a highly challenging task in the community of computational solid mechanics. The well-established and widely applied finite element (FEM) [1,2] and boundary element (BEM) [3,4] methods have long been dominant numerical techniques in solving problems with cracks. Beside the FEM and BEM, various meshless methods have been rapidly improved and can be nowadays viewed as competitive alternative for solving numerically fracture mechanics problems. The methods include, but are not limited to, the meshless manifold method [5], the element-free Galerkin method (EFGM) [6], the meshless local Petro-Galerkin method (MLPG) [7,8], methods based on radial basis functions (RBFs) [9–15], and the method of fundamental solutions (MFS) [16–22].

The generalized finite difference method (GFDM) belongs to the family of meshless collocation method and has been proved to be an efficient and accurate method in solving a wide range of boundary value problems [23–29]. Although impressive results have been achieved, no applications of the method to dynamic crack analysis have been reported in literature so far, to the best of the authors' knowledge. In this paper we propose a new framework for dynamic crack analysis by using the generalized finite difference method (GFDM). A self-adaptive strategy is proposed in order to appropriately describe the local asymptotic behavior of the near-tip solutions. The present strategy is to keep the subdomains fixed while increasing automatically the degree of the Taylor series in the regions near the crack-tips (similar to the *p*-refinement strategy used in the FEM). Two different time-stepping schemes, namely the "Houbolt" and

* Corresponding author.
E-mail address: 2021020254@qdu.edu.cn (B. Ju).

https://doi.org/10.1016/j.rinam.2023.100418
Received 21 October 2023; Received in revised form 27 November 2023; Accepted 29 November 2023
Available online 10 December 2023
2590-0374/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

"Newmark" methods, in conjunction with the present GFDM are applied for solving the space-time boundary value problem numerically. Several benchmark numerical examples are presented to demonstrate the accuracy of the present method. Numerical results obtained by using the BEM are also given for the purpose comparison.

2. Problem statement and some foundations of elastodynamics

The basic equations of linear elastodynamics can be expressed as follows:

$$\sigma_{ij,j} + f_i = \rho \ddot{u}_i, \quad i = 1, 2, \tag{1}$$

with the following initial and boundary conditions:

$$u_i(x, y, t) = \dot{u}_i(x, y, t) = 0, \quad for\ t \le 0, \tag{2}$$

$$p_i = \sigma_{ij}(x, y, t) n_j = 0, \quad x \in \Gamma_c^+ \cup \Gamma_c^-, \tag{3}$$

$$p_i = \bar{p}_i, \quad x \in \Gamma_\sigma, \quad u_i = \bar{u}_i, \quad x \in \Gamma_u, \tag{4}$$

where σ_{ij}, u_i and f_i stand for the stress, displacement and body force components, respectively, ρ is the mass density, dots above u_i denote derivatives with respect to time t, i.e., $\ddot{u}_i = \partial^2 u_i / \partial t^2$, $p_i = \sigma_{ij} n_j$ is the traction component with n_j being the unit outward normal vector, $\Gamma_c^+ \cup \Gamma_c^-$ stands for the upper and the lower crack-surfaces where the boundary conditions are, in general, traction-free, Γ_σ and Γ_u represent the remaining boundaries where the traction and displacement conditions are given, respectively. The stresses and strains are related by Hooke's law as follows:

$$\sigma_{ij} = \lambda \varepsilon_{kk} \delta_{ij} + 2\mu \varepsilon_{ij}, \quad \lambda = E\nu / (1+\nu)(1-2\nu), \quad \mu = E / 2(1+\nu), \tag{5}$$

where δ_{ij} represents the Kronecker delta, λ and μ are the so-called Lamé constants, E and ν are Young's modulus and Poisson's ratio, respectively. The strains ε_{ij} can be expressed as:

$$\varepsilon_{ij} = \left(u_{i,j} + u_{j,i}\right) / 2. \tag{6}$$

Inserting Eqs. (5) and (6) into Eq. (1) yields the well-known *Navier-Lamé equations*:

$$(\lambda + \mu) u_{j,ji} + \mu u_{i,jj} + f_i = \rho \ddot{u}_i, \quad i = 1, 2. \tag{7}$$

3. The generalized finite difference method (GFDM) for dynamic crack analysis

3.1. Review of the meshless GFDM formulation

Suppose $\{x_1, \ldots, x_N\}$ being a discretization of the computational domain Ω with N points. For each single point inside $\{x_1, \ldots, x_N\}$, referred to as the *central point* $x_0 = (x_0, y_0)$, the M nearest points $x_i = (x_i, y_i)$ around it should be found. Then the local subdomain Ω_s is defined as the area that contains all these $M+1$ points. Suppose that $U_0 = U(x_0)$ and $U_i = U(x_i)$ being the values of unknown functions at points x_0 and $x_i \in \Omega_s$, respectively. The Taylor series approximation for $U(x_i)$ about point x_0 can be written as:

$$\begin{aligned} U(x_i) = U(x_0) &+ \left(h_i \frac{\partial}{\partial x} + k_i \frac{\partial}{\partial y}\right) U(x_0) \\ &+ \frac{1}{2!}\left(h_i \frac{\partial}{\partial x} + k_i \frac{\partial}{\partial y}\right)^2 U(x_0) + \cdots + \frac{1}{n!}\left(h_i \frac{\partial}{\partial x} + k_i \frac{\partial}{\partial y}\right)^n U(x_0) + \cdots, \quad i = 1, 2, \ldots, M, \end{aligned} \tag{8}$$

where $h_i = x_i - x_0$ and $k_i = y_i - y_0$. Let's suppose that the Taylor series expansions (8) are truncated after, for example, the 4th-order derivatives, the following series truncation error can then be obtained:

$$B(U) = \sum_{i=1}^{M} \left\{ \left[U(x_0) - U(x_i) + \left(h_i \frac{\partial}{\partial x} + k_i \frac{\partial}{\partial y}\right) U(x_0) + \cdots + \frac{1}{4!}\left(h_i \frac{\partial}{\partial x} + k_i \frac{\partial}{\partial y}\right)^4 U(x_0) \right] \omega_i \right\}^2, \tag{9}$$

where ω_i is the weighting function [27]. Adopting notations $U_0 = U(x_0)$, ... , $U_{0,xyyy} = \partial^4 U_0 / \partial x \partial y^3$ and minimizing the error B with respect to all the partial derivatives $U_{0,x}$, ... , $U_{0,xyyy}$ leads to:

$$\begin{cases} \dfrac{\partial B(U)}{\partial U_{0,x}} = \sum_{i=1}^{M} \left[\Phi(U_0,\ U_i) h_i \omega_i^2\right] = 0, \\ \dfrac{\partial B(U)}{\partial U_{0,y}} = \sum_{i=1}^{M} \left[\Phi(U_0,\ U_i) k_i \omega_i^2\right] = 0, \\ \dfrac{\partial B(U)}{\partial U_{0,xx}} = \sum_{i=1}^{M} \left[\Phi(U_0,\ U_i) \dfrac{h_i^2}{2} \omega_i^2\right] = 0, \ \ldots\ldots \end{cases} \tag{10}$$

where $\Phi(U_0,\ U_i) = U_0 - U_i + h_i U_{0,x} + \cdots$. For 4th-order Taylor series expansions, this gives a set of 14 linear equations with 14 unknowns $U_{0,x},\ \ldots,\ U_{0,xyyy}$. The above linear system of equations may be written in the following matrix form:

$$A_{14\times 14} \begin{bmatrix} U_{0,x} \\ U_{0,y} \\ U_{0,xx} \\ U_{0,yy} \\ \cdots \end{bmatrix}_{14\times 1} = B_{14\times(M+1)} \begin{pmatrix} U_0 \\ U_1 \\ \vdots \\ U_M \end{pmatrix}_{(M+1)\times 1}, \tag{11}$$

where $A_{14\times 14}$ and $B_{14\times 14}$ are matrixes with respect to h_i, k_i and ω_i. Premultiplying both sides of Eq. (11) by A^{-1}, leads to:

$$\begin{bmatrix} U_{0,x} \\ U_{0,y} \\ U_{0,xx} \\ U_{0,yy} \\ \cdots \end{bmatrix}_{14\times 1} = A^{-1} B \begin{pmatrix} U_0 \\ U_1 \\ \vdots \\ U_M \end{pmatrix}_{(M+1)\times 1} = \begin{pmatrix} \sum_{i=0}^{M} \lambda_i^{(x)} U_i \\ \sum_{i=0}^{M} \lambda_i^{(y)} U_i \\ \sum_{i=0}^{M} \lambda_i^{(xx)} U_i \\ \cdots \end{pmatrix}, \tag{12}$$

and it can be shown that the derivatives of the unknown function $U(x_0)$ can now be approximated by a linear combination of function values at points inside the local subdomain Ω_s, i.e., $U_{0,x} = \sum_{i=0}^{M} \lambda_i^{(x)} U_i$. By repeating the same procedures for every point scattered inside the domain and substituting the relations (12) into the governing equation and the corresponding boundary conditions, the final linear system of equations can be obtained. On solving this system of equations, the unknown quantities at any point inside the computational domain can be calculated. In order to accurately describe the singular behavior of the near-tip solutions, the 4th-order Taylor series expansions are adopted in the near-tip regions while the 2nd-order expansions are used in the regions far away from the crack-tips. Two implicit time-stepping algorithms, namely the "Houbolt" and "Newmark" methods, are applied for solving the time-dependent Eq. (7) numerically.

3.2. Calculation of dynamic stress intensity factors

For cracked bodies, the displacements are discontinuous across the crack-faces. One direct way for removing such discontinuity is the use of the sub-domain technique. For this purpose the cracked domain should be first divided into two independent sub-domains along the crack-faces and an artificial interfaces (see Fig. 1(a)). The GFDM is then applied separately to each sub-domain to form a linear system of equations like Eq. (7). Finally, by using the displacement continuity and traction equilibrium conditions on the artificial interfaces, the equations written for each sub-domain can be coupled together and the unknowns at any points inside the computational domain can be obtained.

Once the displacements, stresses and strains have been computed by using the schemes described above, the Mode-I and/or Mode-II

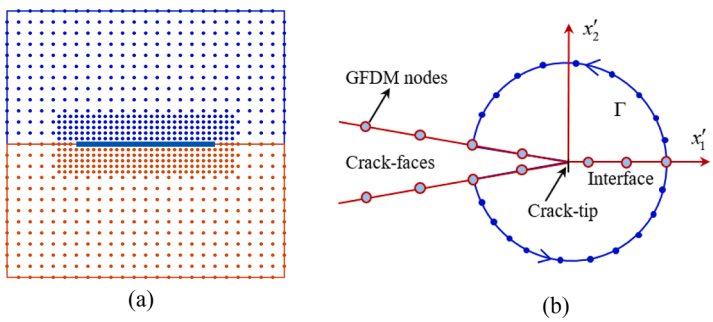

Fig. 1. Profiles of the cracked body (a) and the regions near the crack-tip (b).

dynamic stress intensity factors can be obtained by using the following path independent J-integral method:

$$J = \oint_{\Gamma} \left(W n_1 - \sigma_{ik} n_k \frac{\partial u_i}{\partial x} \right) ds, \tag{13}$$

where Γ is an arbitrary closed surface around the crack-tip with the outward unit normal vector n_k (see Fig. 1(b)), W denotes the strain energy density. For linear elastic materials, the strain energy density can be calculated as $W = \sigma_{ij}\varepsilon_{ij}/2$. Once the J-integral has been calculated, the Mode-I, Mode-II, and/or Mode-III dynamic SIFs can be obtained by using the following relation:

$$J = \frac{1}{E'}\left(K_I^2 + K_{II}^2\right) + \frac{1}{2\mu}K_{III}^2, \tag{14}$$

where $E' = E$ for plane stress and $E' = E/(1-\nu^2)$ for plane strain. Here K_I, K_{II} and K_{III} denote the pure Mode-I, Mode-II and Mode-III SIFs, respectively.

4. Numerical results and discussions

All numerical results are carried out here using MATLAB 2021b, running on a computer with i7 2.90 GHz CPU, 32GB memory, and a Windows 11 operating system. The relative error is defined by:

$$Error = \sqrt{\sum_{k=1}^{N}\left(I_k^{numerical} - I_k^{exact}\right)^2} \Big/ \sqrt{\sum_{k=1}^{N}\left(I_k^{exact}\right)^2}, \tag{15}$$

where $I_k^{numerical}$ and I_k^{exact} denote the numerical and exact solutions at the kth point, respectively.

4.1. A rectangular plate with a central crack of length 2a

As shown in Fig. 2, we consider a rectangular plate with a central crack of length $2a$. The cracked plate is subjected to a uniform impact loading on its upper and lower boundaries. This example has been studied by Zhang and Savaidis by using an advanced time-domain BEM in Ref. [30]. A total number of 6756 nodes have been used for discretizing the computational domain. The nodal-mesh close to the near-tip regions is refined with the smallest mesh-size in the vicinity of the crack-tip is taken as 0.2 mm. The time-step size is chosen as $\Delta t = 0.1635\mu s$. The time-stepping scheme used in this example is Newmark's method with $\delta = 1/2$ and $\alpha = 1/4$.

Normalized dynamic SIFs $\bar{K}_I = K_I/(\sigma(t)\sqrt{\pi a})$ versus the time t obtained by using the proposed method are shown in Fig. 3. Numerical results obtained by Zhang and Savaidis [30] via an advanced time-domain BEM and by Chen [31] via a finite difference method are also given for the purpose of comparison. It can be observed that, by compassion, the proposed method provides very satisfactory numerical results. The discrepancy between the numerical results could be caused by different time-steps and techniques employed for dynamic analysis. The colored contour plots for estimated stresses σ_{22} at times $t = 4.579\mu s$ and $t = 10.793\mu s$ are shown in Fig. 4 which clearly depicts the crack-tip singularities. It is reported that the time used to perform the simulations is 56.34 s. The convergence analysis of the GFDM for solving crack problems can be found in Ref. [32].

4.2. An edge-cracked plate subject to variable impact loadings

A rectangular plate with an edge crack of length a subject to variable impact loadings is studied (see Fig. 5). The material constants,

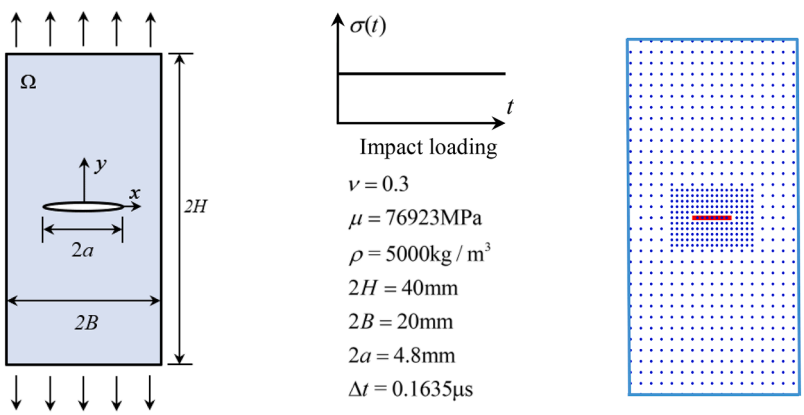

Fig. 2. A center-cracked plate subjected to remote impact loading (plane strain).

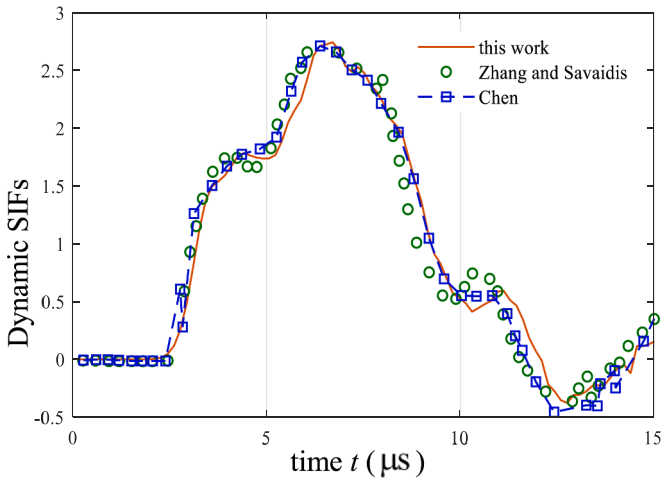

Fig. 3. Normalized dynamic stress intensity factors.

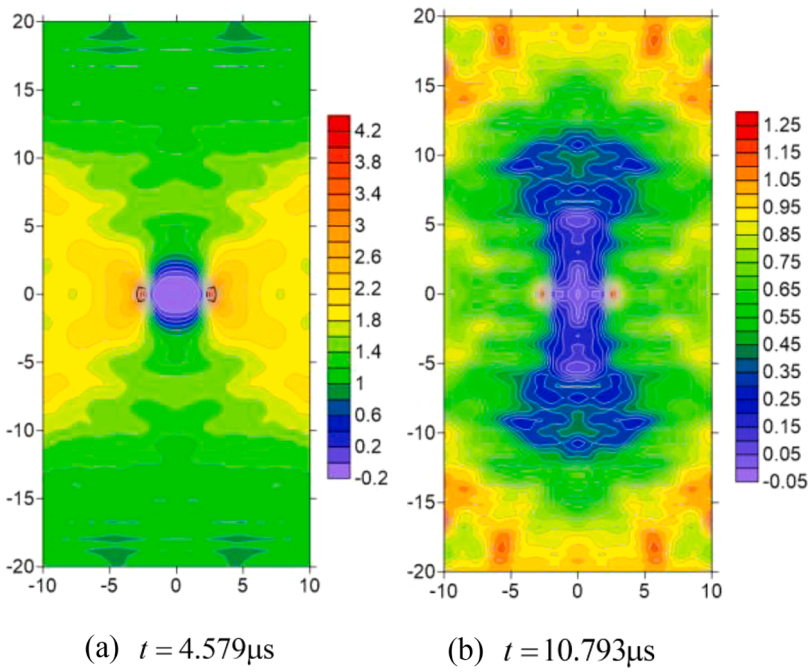

Fig. 4. The colored contour plots for stresses (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

geometry data and the time-stepping algorithm used here are identical to those in the previous example. Numerical results for normalized dynamic SIFs are shown in Fig. 6. Results provided by Zhang [33] via the BEM are also given for the purpose of comparison where, again, both results are pretty agreement with each other at each time-steps. Finally, the colored contour plot for estimated stresses σ_{22} at time $t = 19\mu s$ are shown in Fig. 7.

5. Concluding remarks

This paper documents the first attempt to apply the generalized finite difference method (GFDM) for transient elastodynamic crack analysis. The proposed method is truly meshless which means that no mesh, but a mere collocation of points is required. Therefore, the burdensome remeshing procedures associated with the traditional mesh-based method can be avoided. Two different time-stepping schemes are employed for solving the resulting time-dependent equations. The path-independent J-integral and sub-domain technique are adopted to compute the dynamic stress intensity factors. Our preliminary numerical analysis show that the proposed method

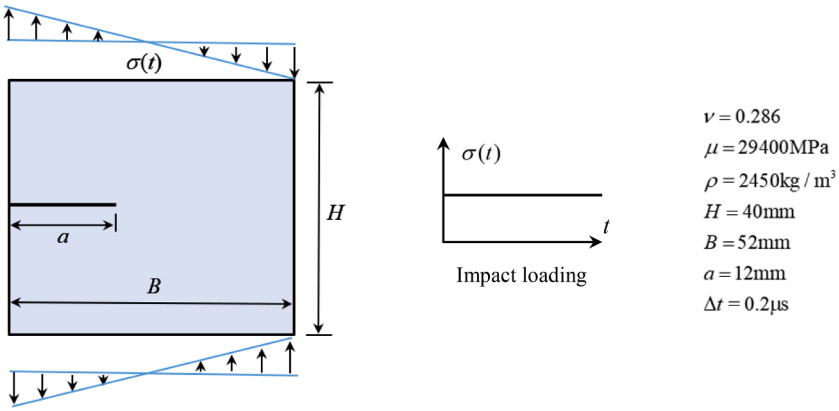

Fig. 5. An edge-cracked plate subject to uniform impact loading.

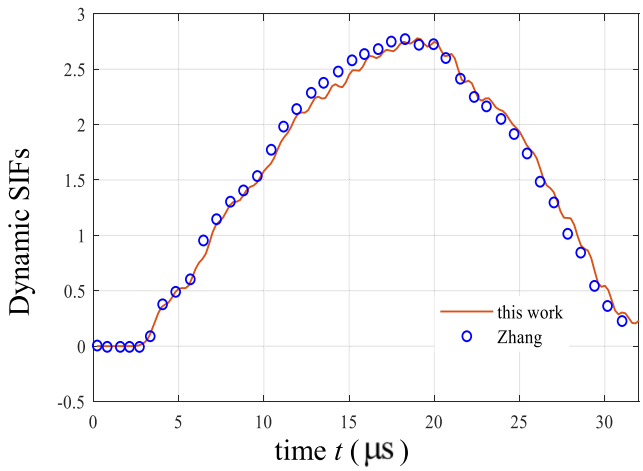

Fig. 6. The normalized dynamic stress intensity factors.

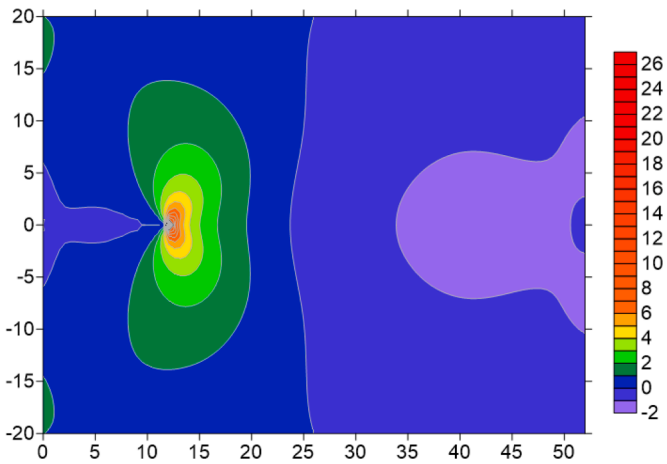

Fig. 7. The colored contour plots for stresses.

is very promising for dynamic crack analysis. Although the method is developed in the context of two-dimensional/in-plane crack problems, extension of the method to, for example, the interface cracks in composite materials as well as three-dimensional dynamic crack problems, is fairly straightforward. The present study provides a solid basis for these interesting topics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Barsoum RS. On the use of isoparametric finite elements in linear fracture mechanics. Int J Numer Method Eng 1976;10(1):25–37.
[2] Zou M-S, Tang H-C, Liu S-X. Modeling and calculation of acoustic radiation of underwater stiffened cylindrical shells treated with local damping. Marine Struct 2023;88:103366.
[3] Gu Y, Zhang C. Fracture analysis of ultra-thin coating/substrate structures with interface cracks. Int J Solids Struct 2021;225:111074.
[4] Gu Y, Zhang C. Novel special crack-tip elements for interface crack analysis by an efficient boundary element method. Eng Fract Mech 2020;239:107302.
[5] Li SC, Li SC, Cheng YM. Enriched meshless manifold method for two-dimensional crack modeling. Theoret Appl Fract Mech 2005;44(3):234–48.
[6] Chen S, Liu Y, Cen Z. Lower bound shakedown analysis by using the element free Galerkin method and non-linear programming. Comput Method Appl Mech Eng 2008;197(45):3911–21.
[7] Atluri SN, Zhu T. A new Meshless Local Petrov-Galerkin (MLPG) approach in computational mechanics. Comput Mech 1998;22(2):117–27.
[8] Li X, Dong H. An element-free Galerkin method for the obstacle problem. Appl Math Lett 2021;112:106724.
[9] Zheng H, Zhang C, Wang Y, Sladek J, Sladek V. A meshfree local RBF collocation method for anti-plane transverse elastic wave propagation analysis in 2D phononic crystals. J Comput Phys 2016;305:997–1014.
[10] Gu Y, Fan C-M, Fu Z. Localized method of fundamental solutions for three-dimensional elasticity problems: theory. Adv Appl Math Mech 2021;13(6):1520–34.
[11] Ju B, Qu W. Three-dimensional application of the meshless generalized finite difference method for solving the extended Fisher–Kolmogorov equation. Appl Math Lett 2023;136:108458.
[12] Zhao S, Gu Y. A localized Fourier collocation method for solving high-order partial differential equations. Appl Math Lett 2023;141:108615.
[13] Fu Z, Tang Z, Xi Q, Liu Q, Gu Y, Wang F. Localized collocation schemes and their applications. Acta Mechanica Sinica 2022;38(7):422167.
[14] Fu Z-J, Xie Z-Y, Ji S-Y, Tsai C-C, Li A-L. Meshless generalized finite difference method for water wave interactions with multiple-bottom-seated-cylinder-array structures. Ocean Eng 2020;195:106736.
[15] Xi Q, Fu Z, Zou M, Zhang C. An efficient hybrid collocation scheme for vibro-acoustic analysis of the underwater functionally graded structures in the shallow ocean. Comput Method Appl Mech Eng 2024;418:116537.
[16] Qu W, Gao H, Gu Y. Integrating Krylov deferred correction and generalized finite difference methods for dynamic simulations of wave propagation phenomena in long-time intervals. Adv Appl Math Mech 2021;13:1398–417.
[17] Chen CS, Fan CM, Wen PH. The method of approximate particular solutions for solving certain partial differential equations. Numer Method Part Differen Eq 2012;28(2):506–22.
[18] Ma J, Chen W, Lin J. Crack analysis by using the enriched singular boundary method. Eng Anal Bound Elem 2016;72:55–64.
[19] Chen Z, Sun L. A boundary meshless method for dynamic coupled thermoelasticity problems. Appl Math Lett 2022;134:108305.
[20] Wang X, Gu Y, Golub MV. Analysis of bimaterial interface cracks using the localized method of fundamental solutions. Result Appl Math 2022;13:100231.
[21] Gu Y, Fan C-M, Xu R-P. Localized method of fundamental solutions for large-scale modeling of two-dimensional elasticity problems. Appl Math Lett 2019;93:8–14.
[22] Gu Y, Lin J, Fan C-M. Electroelastic analysis of two-dimensional piezoelectric structures by the localized method of fundamental solutions. Adv Appl Math Mech 2023;15:880–900.
[23] Fan CM, Huang YK, Li PW, Chiu CL. Application of the generlized finite-difference method to inverse biharmonic boundary-value problems. Numer Heat Tranf. B-Fundam. 2014;65(2):129–54.
[24] Fu Z-J, Tang Z-C, Zhao H-T, Li P-W, Rabczuk T. Numerical solutions of the coupled unsteady nonlinear convection-diffusion equations based on generalized finite difference method. Eur Phys J Plus 2019;134(6):272.
[25] Qu W, He H. A spatial–temporal GFDM with an additional condition for transient heat conduction analysis of FGMs. Appl Math Lett 2020;110:106579.
[26] Li P-W, Fu Z-J, Gu Y, Song L. The generalized finite difference method for the inverse Cauchy problem in two-dimensional isotropic linear elasticity. Int J Solid Struct 2019;174-175:69–84.
[27] Gu Y, Qu W, Chen W, Song L, Zhang C. The generalized finite difference method for long-time dynamic modeling of three-dimensional coupled thermoelasticity problems. J Comput Phys 2019;384:42–59.
[28] Jiang S, Gu Y, Golub MV. An efficient meshless method for bimaterial interface cracks in 2D thin-layered coating structures. Appl Math Lett 2022;131:108080.
[29] Xia H, Gu Y. Generalized finite difference method for electroelastic analysis of three-dimensional piezoelectric structures. Appl Math Lett 2021;117:107084.
[30] Zhang C, Savaidis A. Time-domain BEM for dynamic crack analysis. Math Comput Simul 1999;50(1–4):351–62.
[31] Chen YM. Numerical computation of dynamic stress intensity factors by a Lagrangian finite-difference method (the HEMP code). Eng Fract Mech 1975;7(4):653–60.
[32] Lei J, Xu Y, Gu Y, Fan C-M. The generalized finite difference method for in-plane crack problems. Eng Anal Bound Elem 2019;98:147–56.
[33] Zhang C. A 2D hypersingular time-domain traction BEM for transient elastodynamic crack analysis. Wave Motion 2002;35(1):17–40.